

Name

- For the ODE $y'' + 2y' - 8y = t$

- Write down the complementary sol.

$y_c(t) =$

- Compute a particular sol.

$y_p(t) =$

- Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$

1. For the ODE $y'' + 4y' + 13y = e^t$

- 1.1. Write down the complementary sol.

$y_c(t) =$

- 1.2. Compute a particular sol.

$y_p(t) =$

- 1.3. Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$

- For the ODE $2y'' - 4y' + 20y = 40e^t$ with

- Write down the complementary sol.

$$y_c(t) =$$

- Compute a particular sol.

$$y_p(t) =$$

- Describe the solution as $t \rightarrow \infty$

$$\text{As } t \rightarrow \infty$$

2. For the ODE $y'' + 2y' + 5y = 17e^{-t/2}$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Compute a particular sol.

$$y_p(t) =$$

- 2.3. Describe the solution as $t \rightarrow \infty$

$$\text{As } t \rightarrow \infty$$